Niladri Talukder

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PROFILE:

Ph.D. Mechanical Engineer with 6+ years of experience in functional materials and hybrid composites development, and R&D focused research. Expertise in materials formulation, synthesis, experimental design, and performance evaluation using microscopy, spectroscopy, thermal analysis, and electrochemical measurements. Experienced in corelating material chemistry, interfaces, degradation, and processing–structure–property relationships to lead materials and process optimization. Skilled in leading experimental projects from concept and sample preparation through testing, troubleshooting, data interpretation, and technical reporting. Proven record of collaboration with academic researchers, National Laboratory scientists, and industry partners; mentoring researchers, and translating research findings into gradual improvements of materials; publications and supporting commercialization initiatives of in-house technologies.

EDUCATION:

Degree	***Institution***	***Duration***
Ph.D. in Mechanical Engineering	New Jersey Institute of Technology, United States	Sep 2020—Dec 2025
M.Sc. in Mechanical Engineering	Andong National University, South Korea	Sep 2015—Aug 2017
B.Sc. in Mechanical Engineering	Khulna University of Engineering & Technology, Bangladesh	Jan 2009—Sep 2013

PROFESSIONAL EXPERIENCES:

Data Analyst (Research)

New Jersey Institute of Technology, Newark, NJ, USA. *Oct 2026 — Present*

- Conducting research on graphene-based nanomaterials, MXenes, and MOFs, including material synthesis, process optimization, characterization, experimental testing, and performance validation for various engineering applications.
- Developing and evaluating functional nanomaterials and hybrid composites for electrocatalytic systems and phase change materials (PCMs); Performing comprehensive physical, chemical, electrochemical, and thermal characterization to investigate material properties, performance, long-term stability, and further improvement of materials.
- Coordinating multiple research projects in collaboration with laboratory members and external collaborators, supporting experimental planning, data analysis, and technical documentation; Developing and evaluating technical specifications for laboratory equipment, components, and materials procurement.

Scientist

Abonics, Inc., Tenafly, NJ, USA. *May 2026 — Present*

- Leading research and technology innovation; materials and prototype development for biochips; Grant proposals writing; Contributing to tasks related to securing intellectual properties, and external collaborations.
- Supporting the efforts of transition from prototype to scalable product development; Ensuring quality standards; Supporting technical tasks for relevant regulatory compliances; Activity supporting product commercialization efforts.

Research and Teaching Assistant

New Jersey Institute of Technology, Newark, NJ, USA. *Sep 2020 — Jan 2026*

- Project Lead: *Carbon-based nanomaterials for electrochemical energy storage and conversion systems*.

  Synthesize nitrogen-doped graphene/ZIF-8 (N-G/MOF) nanocatalysts for the oxygen reduction reaction (ORR) via ball milling and thermal treatment; optimize N-doping ratio and active sites by controlling milling speed and time, achieving catalytic performance surpassing N-G and approaching benchmark 10 wt% Pt/C catalyst performance.

  Evaluate long-term stability and degradation mechanisms of N-G/MOF nanocatalysts under ORR conditions using electrochemical and advanced structural characterization techniques, establishing relationships between physical and chemical structures and catalytic performance under different operational environments.

- Project Lead: *Nanomaterials-supported Phase Change Materials (PCMs) for thermal property enhancement*.

  Leverage molecular-level interactions of engineered nanomaterials and PCMs to enhance thermal performance. Achieved ~5% increase in latent heat and ~10% improvement in thermal stability relative to the base PCM.

Lead a three-member team to assess the commercialization potential of Nano-PCM composites through the NSF I-Corps program. Conducted customer discovery interviews for market analysis, and value proposition validation.

- Project Lead: *AI-based Functionality Prediction of Carbon-based Nanomaterials for Energy Applications*.

  Implementation of Artificial Intelligence (AI) to the functionalities of carbon-based nanomaterials in two crucial energy applications: (1) as electrocatalysts for oxygen reduction reactions (ORR) in electrochemical energy systems and (2) as support materials for phase change materials (PCMs) in thermal energy storage systems.

- Project Lead: *Integrated Solar Photovoltaic and Heat Pump System for Coupled Thermal and Electrical Energy Utilization*.

  Evaluate the potential of capturing the thermal energy generated by PV panels and redirect it to support household heat pump systems to improve overall energy utilization. Coupling photovoltaic electricity generation with thermal energy recovery to develop an efficient hybrid renewable energy system for domestic applications.

- Project Co-lead: *Microfluidic-based multiplex assay biochip development for point-of-care disease detection*.

  Developed capacitance-based biosensors integrated with microfluidic biochips for rapid detection of ovarian cancer biomarkers CA-125 and HE4 and Alzheimer's disease biomarkers Aβ42/Aβ40. Achieved biomarker detection in less than 40 seconds with high sensitivity and specificity. Ran commercialization efforts with this technology.

- Led six experimental research projects on the development of functional nanomaterials, emphasizing structure–property relationships for electrochemical fuel cells, batteries, and thermal energy storage materials at the Center for Functional Nanomaterials (CFN), Brookhaven National Laboratory.
- Maintained safe laboratory operations; oversaw equipment and materials procurement, storage, and maintenance; ensured compliance with relevant regulatory process; managed coordination between experimenters and schedules.
- Developed experimental protocols; ensured reliable experiment executions, data management, and documentation.
- Mentored 7+ undergraduate and graduate researcher assistants in materials synthesis, electrochemical testing, and data analysis; oversaw concurrent experimental activities, timely technical outcomes, and project reports preparation.
- Served as Teaching Assistant for 12 undergraduate and graduate engineering courses at NJIT for 4 years; maintained clear technical communication, delivered lectures and laboratory instructions, monitored students' progress.

Executive (Engineering)

Source Associate Ltd., Dhaka, Bangladesh. *Nov 2019 — Dec 2020*

- Oversaw plant erection and installation activities to ensure compliance with NFPA standards; managed material planning and control processes; and developed detailed project plans and Bills of Quantities (BOQ).

Graduate Research Assistant

Andong National University, Andong, South Korea. *Sep 2015 — Oct 2017*

- Led projects for designing and operating a constant-volume combustion chamber with Schlieren and shadowgraph diagnostics for high-precision combustion studies. Measured laminar flame speed and Markstein length of alternative fuels, including n-butanol and fatty acid methyl esters (FAME). Combustion models supported by experimental data.

AREAS OF RESEARCH INTERESTS:

Functional Nanomaterials Synthesis and Processing; Graphene-Based Nanomaterials and Hybrid Composites; Advanced Materials Characterization; Electron Microscopy (SEM/HR-TEM); Nanoscale Structure–Property Relationships; Surface Chemistry and Defect Analysis; Materials Stability and Degradation; Electrochemical Energy Systems; Phase Change Materials (PCMs); Thermal Energy Storage Systems; Materials Discovery and Applications.

SKILS & EXPERTIES:

Advanced Materials and Synthesis: Graphene, N-doped graphene, MXene, MOFs, and nanomaterial-enhanced PCM composites; formulation, synthesis, functionalization, dispersion, and processing–structure–property optimization.

Materials Characterization and Analytical Methods: SEM, FTIR, XPS, Raman, TEM, EDX, XRD, and thermal analysis (DSC, TGA); analysis of composition, morphology, interfaces, microstructure, operational stability for materials development.

Degradation and Root-Cause Investigation: Investigation of surface chemistry, interfacial changes, performance loss, degradation mechanisms; integration of analytical results to identify factors contributing to materials' performance.

Technical Project Development and Execution: Planning and execution of materials analysis projects, including sample preparation, instrument setup, method selection, troubleshooting, data interpretation, and final technical reporting.

Experimental Design and Performance Validation: Comparative experimental studies, development and optimization of methods, evaluation of thermal and chemical stability, and data interpretation to guide process improvements.

Engineering Tools and Data Analysis: Interpretation of complex experimental datasets, trend and variability analysis, data visualization, and evidence-based engineering recommendations; proficient in MATLAB, Python, Origin, CAD, and C.

Laboratory Operations and Technical Documentation: Organized sample handling, analytical instrumentation workflows, laboratory safety, process optimization, SOP preparation, experimental procedures, and technical reports.

Technical Leadership and Collaboration: Mentoring junior researchers; collaborating with scientists, engineers, national laboratory researchers, and industry partners; explaining analytical results through clear reports and presentations.

ACHIEVEMENTS, AWARDS, and SERVICES:

- US Patent: Phase Change Material Composites Including Carbon Nanomaterials. Non-provision Patent Application, **July 2025** (US20260028517A1) Inventors: Lee, E. S. & **Talukder, N.**
- National Science Foundation Innovation-Corps Award, National Cohort 2025. I-Corps: Translation Potential of Nano-carbon-enhanced Optimizable Phase Change Materials (Award #2528474). **Talukder, N.** (Entrepreneurial Lead: EL).
- National Science Foundation Innovation-Corps Award, Northeast Regional Program 2024. **Talukder, N.** (EL).
- National Science Foundation Innovation-Corps Award, National Cohort 2023. I-Corps: Multiplex Diagnostic Assay Using Interdigitated Nano-sensing Technology Implemented Point-of-Care Device (Award # 2318433). **Talukder, N.** (Co-EL).
- National Science Foundation Innovation-Corps Award, Northeast Regional Program 2022. **Talukder, N.** (Co-EL).
- Served as a Key Personnel in an NSF-funded Technology Innovation Translation and Acceleration (TITA) project at NJIT.
- Served as Reviewer for multiple renowned peer-reviewed Journals published by ELSEVIER, RCS, and MDPI.
- Selected for Invited Graduate Student Talk in the New York Metropolitan Catalyst Society (NYCS) Meeting February Meeting 2024 at the Princton University.
- Delivered 15+ oral presentations and talks in academic research and professional engineering conferences/meetings.
- Wrote 6 successful general user grant proposals for facilities access at Brookhaven National Laboratory.
- Prepared 2 successful academia-industry collaborative research proposals and work-flow documents.
- Invited for Guest Lecture at the Material Advantage Society Meetings, NJIT Chapter, August 2023 and March 2024.

PUBLICATIONS & PRESENTATIONS:

Book Monograph

Lee, E. S. & **Talukder, N.** Nitrogen-doped Graphene Nanomaterials for Electrochemical Systems – Fundamentals and Applications. *Springer Nature*, **June 2026**. *[Link]*

Peer-reviewed Journal Articles:

1. **Talukder, N.**; Wang, Y.; Tong, X.; Lee, E. S. Chemical Changes from N-doped Graphene and Metal-organic Frameworks to N-G/MOF Composites for Improved Electrocatalytic Activity. *Carbon* **2025**, 232, 119816. [Link]
2. Mannaf, M. M.; Rahman, M. M.; Sabuj, S. T.; **Talukder, N.**; Lee, E. S. Current Progress in Advanced Functional Membranes for Water-Pollutant Removal: A Critical Review. *Membranes* **2025**, 15, 300. [Link]
3. Wang, Y.; **Talukder, N.**; … Tong, X.; Lu, M.; Lee, E.S. Enhanced Stability and Sensitivity for CA-125 Detection Under Microfluidic Shear Flow Using Polyethylene Glycol-coated Biosensor. *ACS Omega* **2025**, 10 (1), pp. 692-702. [Link]
4. **Talukder, N.**; Wang, Y.; …Tong, X.; Lee, E. S. An Investigation on the Structural Stability of ZIF-8 in Water versus Water-derived Oxidative Species in Aqueous Environment. *Microporous and Mesoporous Materials* **2024**, 366, 112934. [Link]
5. **Talukder, N.**; Wang Y.; Nunna B. B.; Lee E. S. N-doped Graphene (NG)/MOF (ZIF-8)-Based/derived Materials for Electrochemical Energy Applications: Synthesis, Characteristics, and Functionality. *Batteries* **2024**, 10 (2), 47. [Link]
6. Wang, Y.; **Talukder, N.**; Nunna, B. B.; and Lee, E. S. Dean Vortex-enhanced Blood Plasma Separation in Self-driven Spiral Microchannel Flow with Cross-flow Microfilters. *Biomicrofluidics* **2024**, 18 (1). 014104. [Link]
7. **Talukder, N.**; Wang, Y.; Nunna, B. B.; Tong, X.; Boscoboinik, A. J.; Lee, E. S. Investigation on Electrocatalytic Performance and Material Degradation of N-doped Graphene-MOF Nanocatalyst in Emulated Electrochemical Environments. *Industrial Chemistry & Materials* **2023**, 1, 360−375. [Link]
8. **Talukder, N.**; … Nunna, B. B.; Lee, E. S. An In-Depth Exploration of the Electrochemical Oxygen Reduction Reaction

(ORR) Phenomenon on Carbon-Based Catalysts in Alkaline and Acidic Mediums. *Catalysts* **2022**, 12 (7), 791. [*Link*]

9. Wang, Y.; Nunna, B. B.; **Talukder, N.**; Lee, E. S. Microfluidic-Based Novel Optical Quantification of Red Blood Cell Concentration in Blood Flow. *Bioengineering* **2022**, 9 (6), 247. [*Link*]
10. **Talukder, N.**; Wang, Y.; Nunna, B. B.; Lee, E. S. Nitrogen-doped Graphene Nanomaterials for Electrochemical Catalysis/reactions: A Review on Chemical Structures and Stability. *Carbon* **2021**, 185, 198–214. [*Link*]
11. Wang, Y.; Nunna, B. B.; **Talukder, N.**; Etienne, E. E.; Lee, E. S. Blood Plasma Self-separation Technologies During the Self-driven Flow in Microfluidic Platforms. *Bioengineering* **2021**, 8 (7), 94. [*Link*]
12. Etienne, E. E.; Nunna, B. B.; **Talukder, N.**; Wang, Y.; Lee, E. S. Covid-19 Biomarkers and Advanced Sensing Technologies for Point-of-Care (POC) Diagnosis. *Bioengineering* **2021**, 8 (7), 98. [*Link*]
13. **Talukder, N.**; Lee, K. Y. Laminar Flame Speeds and Markstein Lengths of Methyl Decanoate – Air Premixed Flames at Elevated Pressures and Temperatures. *Fuel* **2018**, 234, 1346–1353. [*Link*]
14. **Talukder, N.**; Lee, K. Y. Laminar Flame Speeds for n-butanol/air Mixtures at Elevated Pressures and Temperatures: An Experimental and Numerical Study. *Journal of Mechanical Science and Technology* **2018**, 32, 1827–1834. [*Link*]
15. **Talukder, N.**; Lee, K. Y. Laminar Flame Speeds for n-butanol/air Mixtures at Elevated Pressures and Temperatures: An Experimental and Numerical Study. *Journal of Mechanical Science and Technology* **2018**, 32, 1827–1834. [*Link*]
16. Rahman, M. M.; … **Talukder, N.**; … Lee E. S. Artificial Intelligence in Hydrogen Energy Systems: A Comprehensive Review of Electrochemical, Storage, Electrolysis, and Combustion Applications. *Next Energy* **2026**. [in press]
17. Yousefi, H.; **Talukder, N.**; Govindan, D.; Zhao, M-Q.; Lee, E. S. Nitrogen-Doped Graphene Stabilizes Partially Etched Ti_3C_2Tx MXene for Durable Oxygen Reduction. *ACS Applied Catalysis* **2026**, [*under review*]

Conference Publications:

1. Wang, Y.; **Talukder, N.**; Cha, K.; Nunna, B. B.; Lee, E. S. Microfluidic-Based Capacitive Biosensor for Multiplex Assay Detection. *IEEE Healthcare Innovations and Point of Care Technologies (HI-POCT)* **September 2024**, pp. 21–24. [*Link*]
2. Nunna, B. B.; Wang, Y.; **Talukder, N.**; Lee, E. S. Capillary Flow Dynamics of Blood with Varied Hematocrit in Microfluidic Platforms. *IEEE Healthcare Innovations and Point of Care Technologies (HI-POCT)* **March 2022**, 1–4. [*Link*]
3. **Talukder, N.**; Lee, K. Y. Experimental Measurement of Laminar Flame Velocity of n-butanol-air Premixed Flame at High Pressure and Temperature. 51st Korean Society of Combustion (KOSCO) Symposium **December 2015**, 87–88. [*Link*]
4. Lee, K. Y.; **Talukder, N.** The Methods Calculating the Reduction Efficiency of Nitrogen Oxide for the Facilities Including the Low NOx Burners. 51st Korean Society of Combustion (KOSCO) Symposium, **December 2015**, 295–296. [Link]

Conference Presentations:

1. Yousefi, H.; Ankush; **Talukder, N.**; Lee, E. S. Partially Etched Ti_3C_2 MXene (from Ti_3AlC_2)/Nitrogen-Doped Graphene Composite as a Low-Cost Electrocatalyst for the Oxygen Reduction Reaction. TMS 2026 Annual Meeting and Exhibition, Orlando, FL, USA, **March 2027**.
2. Yousefi, H.; **Talukder, N.**; Govindan, D.; Zhao, M-Q.; Lee, E. S. Role of Nitrogen-Doped Graphene in Stabilizing Partially Etched Ti_3C_2 MXene during Oxygen Reduction. *MRS Fall Meeting and Exhibit, Boston, MA, USA*, **December 2026**.
3. Lee, E. S.; **Talukder, N.** Thermal Property Enhancement of Phase Change Materials by Engineered 2D N-Doped Graphene Nanoparticle Support. *TMS 2026 Annual Meeting and Exhibition, San Diego, California, USA*, **March 2026**.
4. Jiang, A.; **Talukder, N.**; Lee, E. S. Enhancement of Thermal Functionality in Phase Change Materials Using Nitrogen-Doped Graphene Support Materials. *IMECE, Memphis, TN, USA*, **November 2025**.
5. Lee, E. S.; **Talukder, N**. Advanced ORR Electrocatalyst from Physicochemical Integration of N-doped Graphene, MOF, and MXene by Wet Ball Milling. *TMS 2025 Annual Meeting and Exhibition, Las Vegas, Nevada, USA*, **March 2025**.
6. **Talukder, N.**; Wang, Y.; Nunna, B. B.; Lee, E. S. N-Doped Carbon Nanocatalyst with Zn Single Atom Catalytic Sites from N-Doped Graphene and Metal-Organic Frameworks. *MRS Fall Meeting and Exhibit, Boston, MA, USA*, **December 2024**.
7. Wang, Y.; **Talukder, N.**; Nunna, B. B.; Lee, E. S. Stability Improved Cancer Antigen (CA-125) Detection Using Polyethylene Glycol (PEG) in the Microfluidic Shear Flow Condition. *IEEE EMBC, Orlando, FL, USA*, **July 2024**.
8. **Talukder, N.**; … Lee, E. S. Investigation of the Chemical Changes from N-Doped Graphene and MOF to N-G/MOF Composites for the Enhancement of Electrocatalytic Properties. *245th ECS Meeting, San Francisco, CA, USA*, **May 2024**.
9. **Talukder N.**; Wang Y.; and Lee E.S. Physicochemical Changes for ORR Electrocatalysis Enhancement During Nitrogen doped Graphene with Metal-organic Framework (N-G/MOF) Nanocatalyst Synthesis, *New York Metropolitan Catalyst*

Society (NYCS) 2024 Annual Spring Symposium, Newark, NJ, USA, **March 13, 2024**.

10. **Talukder, N.**; Wang, Y.; Nunna, B. B.; Lee, E. S. Metal-organic Framework Integrated Nitrogen-doped Graphene (N-G/MOF): A High Potential Electrocatalyst for the Oxygen Reduction Reaction. *ACS Spring Meetings and Exposition, Indianapolis, IN, USA*, **March 2023**.
11. **Talukder, N.**; Wang, Y.; Nunna, B. B.; Lee, E. S. On the Hydrolysis of ZIF-8 Crystal Structure: An in-depth Perspective. *ACS Spring Meetings and Exposition Indianapolis, IN, USA*, **March 2023**.
12. Wang, Y.; Nunna, B. B.; **Talukder, N.**; Lee, E. S. Blood Plasma Separation Enhanced by Dean Vortex During the Capillary Flow in the Spiral Microchannel for Point-of-Care (POC) Applications. *ACS Spring 2023 Meetings and Exposition Indianapolis, IN, USA*, **March 2023**.
13. **Talukder, N.**; A. Paliouras, Wang, Y.; Nunna, B. B.; Lee, E. S. Effect of Hydrogen Peroxide (H2O2) on the Graphene-based Electrocatalyst for Oxygen Reduction Reaction (ORR). *IMECE, Columbus, OH, USA*, **October–November 2022**.
14. A. Paliouras, **Talukder, N.**; Wang, Y.; … Lee, E. S. Removal of Impurities from Nitrogen-Doped/Metal Organic Framework Graphene (N-G/MOF) Electrocatalyst for Electrochemical Energy Conversion and Storage Systems. *IMECE®, Columbus, OH, USA*, **October–November 2022**.
15. Wang, Y.; Nunna, B. B.; J. Bae, S. Jadhav, **Talukder, N.**; Lee, E. S. Enhanced Effect of Dean Vortex to Separate Blood Plasma During Capillary Flow in the Spiral Microchannel for Point-of-Care (POC) Biomedical Applications. *IMECE®, Columbus, OH, USA*, **October–November 2022**.
16. Nunna, B. B.; Bae, J.; Wang, Y.; **Talukder, N.**; Lee, E. S. Influence of Hematocrit on the Capillary Blood Flow in PDMS and PMMA Microchannels. *IMECE® International Mechanical Engineering Congress and Exposition®, Columbus, OH, USA*, **October–November 2022**.
17. **Talukder, N.**; Lee, K. Yong. Experimental Investigation on Laminar Flame Speed of Methyl Decanoate (MD)/Air Premixed Flames. *53rd Korean Society of Combustion (KOSCO) Symposium, Jeju, South Korea*, **December 2016**.
18. **Talukder, N.**; Lee, K. Yong. Experimental Measurement of Laminar Flame Speeds of Methyl Decanoate/Air Premixed Flames at Elevated Temperature and Pressures. *52nd Korean Society of Combustion (KOSCO) Symposium, Buyeo, South Korea*, **May 2016**.